

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1004

Outlook 100 kV
December 18, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1004. The Interconnection Customer proposes a 207.1 MW solar photovoltaic facility interconnecting at the Outlook 100 kV substation in the Illinois area. The request was received on May 28, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 337 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,936,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Tower Automotive Substation / Shroud 100.0 kV	Shroud / Tower Automotive 100.0 kV Line 1	121.4	153	22.7
Shroud / Tower Automotive Substation 100.0 kV Line 1	Shroud / Tower Automotive 100.0 kV #2	120.5	154	22.7
Wabash 500/230 kV Transformer Bank	Richland Substation / Dekalb 500.0 kV	106.1	1,022	19.5
St. Lawrence / James De Young 230.0 kV	Lonetree / E Methuen 500.0 kV	104.2	1,621	6.9
Winthrop / Hebo 230.0 kV	Barrett / Hebo 230.0 kV L	100.9	993	5.5
Sabanera Substation / Lonetree 500.0 kV	Muskogee / Montpelier Electric Generating Station 500.0 kV Line	100.9	4,315	10.8
E Methuen / Lonetree 500.0 kV	Muskogee / Montpelier Electric Generating Sta 500.0 kV	100.2	3,951	9.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Shroud / Tower Automotive 100.0 kV	Rebuild line with higher-capacity conductor	\$4,030,000	86% (\$3,460,000)
Shroud / Tower Automotive 100.0 kV #1	Rebuild line with higher-capacity conductor	\$4,699,000	86% (\$4,034,000)
Wabash 500/230kV Auto	Replace transformer with higher-rated unit	\$19,309,000	1% (\$252,000)
James De Young / St. Lawrence 230.0 kV	Reconductor line	\$27,334,000	1% (\$267,000)
Winthrop / Hebo 230.0 kV	Reconductor line	\$4,614,000	1% (\$50,000)
Sabanera / Lonetree 500.0 kV	Reconductor line	\$86,115,000	1% (\$674,000)
Lonetree / E Methuen 500.0 kV	Reconductor line	\$26,398,000	1% (\$199,000)
	Total allocated to project		\$8,936,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.